MILITARY SPECIFICATION

LINK, CARTRIDGE, METALLIC, BELT, CALIBER .50, M15A2

1. SCOPE

1.1 This specification covers one type of metallic belt link intended to hold caliber .50 ammunition in the form of flexible belts for use in a machine gun, caliber .50, Fixed M85.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or-request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	-	Preservation, Methods of.
MIL-W-13855	-	Weapons, Small Arms and Aircraft Armament Subsystems, General Specification for.
MIL-I-45607	-	Inspection Equipment, Acquisition) Maintenance and Disposition of.

STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.

DRAWINGS

U. S.Army Weapons Command

F7791440	-	Link, Cartridge, Metallic Belt, Caliber .50, M15A2.

FSC 1305

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PUBLICATIONS

U. S. Army Weapons Command

P7791440 - Packaging Data Sheet for Link, Cartridge, Metallic Belt, Caliber .50, M15A2.

IEL7791440 - Index of Inspection Equipment List.

(Acceptance Inspection Drawings referenced in this specification form a part of IEL7791440.)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Material

ASTM E18 - Test for Rockwell Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.1). Unless otherwise specified (see 6.1), the first article shall include a pilot pack (see 5.1).

3.2 Material, construction and design. Links shall conform to material, construction and design requirements specified herein, on drawing F7791440 and drawings applicable thereto, and in accordance with the applicable material, construction and design provisions of MIL-W-13855.

3.2.1 Heat treatment. The links shall be heat treated within the hardness range specified on Drawing F7791440.

3.2.1.1 Carburization or decarburization. Total carburization or decarburization is not permissible. Partial carburization or decarburization shall not exceed .001.

3.2.2 Free hinging. Links assembled into 20-round belts shall hinge without binding. Testing shall be in accordance with 4.5.3.5.

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3.2.3 Cartridge linking. Links not previously linked and after application of protective finish as specified on Drawing F779144C shall be capable of allowing cartridges to be fully linked by a force, applied centrally without impact, not exceeding 50 pounds. Testing shall be in accordance with 4.5.3.6.

3.2.4 Cartridge stripping. Links not previously linked and after application of protective finish as specified on Drawing F7991440 shall be capable of allowing cartridges to be fully stripped by a force applied centrally without impact of 50 pounds maximum and 18 pounds minimum. Testing shall be in accordance with 4.5.3.7.

3.2.5 Tab radii. Links shall meet the tab radii requirements as specified on Drawing F7991440. Testing shall be in accordance with 4.5.3.8.

3.2.6 Functioning. Links shall be capable of being assembled into belts and shall operate during the normal firing cycle of the M85 machine gun without malfunctioning, cracking, or breaking. Testing shall be in accordance with 4.6.6.

3.3 Marking. Each link shall be clearly marked in accordance with F7791440 and MIL-W-13855.

3.4 Workmanship. Workmanship shall be in accordance with the workmanship requirements of MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

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4.4 First article inspection. The first article shall be selected from the first inspection lot and submitted for inspection in accordance with contract requirements. The first article shall be representative of the production processes to be used during quality production. The first article links shall be subjected to the quality conformance inspection specified herein, the function test specified in 4.6.6, and such other examinations and tests as are necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection

4.5.1 Inspection lot. Unless otherwise specified, the formation, size and presentation of inspection lots and applicable samples shall be in accordance with MIL-STD-105. Inspection lots shall not exceed 500,000 links per lot. Lots shall be from the same heat of steel and consideration within the limitations of MIL-W-13855.

4.5.1.1 Heat treat lot. The heat treat lot as referenced for hardness testing and carburization or decarburization testing shall be limited to one heat of steel and shall consist of links heat treated in a batch-type furnace at one charge or in a continuous-type furnace during an uninterrupted period extending over not more than one eight-hour shift.

4.5.2 Examination. Examination of links and packaging shall be performed in accordance with the applicable examination provisions of MIL-W-13855 and as specified herein.

4.5.2.1 Classification of defects. Classification of defects and acceptable quality levels (AQL's) for examination in accordance with MIL-STD-105 shall be as specified herein and shall include the following provisions:

a. Unless otherwise specified AQL's apply to each listed characteristic.
b. Where a Government drawing is listed as the method of inspection, the contractor shall use inspection equipment fabricated in accordance with the applicable drawing. In the event the contractor desires to use alternate equipment which equals or exceeds the accuracy requirements of the specified drawing, it shall be submitted to the Government for approval.
c. Where SME (Standard Measuring Equipment) is specified as the method of inspection, the contractor may use any type of industry developed, commercially available, multi-usage equipment or special inspection equipment approved by the Government.
d. Where Visual is specified as the method of dimensional inspection, the characteristics should be either scaled or compared with a specimen of known acceptable quality.

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e. Where Visual is specified as the method of inspection of protective coating, the coating must be complete, uniform in thickness and color, free from pits, corrosion, scratches, and worn or bare spots.

f. Examination for packaging defects specified in 4.5.2.1.2 shall apply to each link or container [illegible] cable.

4.5.2.1.1 Link, cartridge, metallic belt, caliber .50, M15A2 Drawing F7791440

Categories	Defect	AQL	Inspection Method
Critical:	None defined.		
Major			
101	Cracks and tears.	0.25	Visual
102	Improper link contour.	0.25	F11016385
103	Missing or defective protective finish.	0.65	Visual
104	Diameter, connecting loop under.	0.65	F7797433
105	Height location, tab.	0.65	F7797435 D7272611
106	Distance between front and rear loops and connecting loop.	0.65	F7797436
107	Width, connecting loop opening.	0.65	F7797433
108	Height location, tab overhang.	0.65	F11016377 7272905
109	Height location, rear loop diameter to cartridge contact surface.	0.65	D11016340 7272944
110	Length location tab overhang.	0.65	F11016377 D7272905
111	Width, stripper opening.	1.0	F7797433
112	Height, stripper contact surface.	1.0	F7797434
113	Location, stripper contact surface.	1.0	D7797287
114	Length overall.	1.5	F7797433
Minor			
201	Missing or improper marking.	1.5	Visual
202	Improper radii detents.	1.5	Visual
203	Improper radius, bottom of tab overhang.	1.5	B11016854 F7797435
204	Thickness, material.	1.5	C7797154
205	Workmanship.	4.0	Visual

4.5.2.1.2 Packaging. Unless otherwise specified in each defect, the packaging requirements are specified on Packaging Data Sheet P7791440.

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Categories	Defect	AQL
Critical:	None defined.	
Major		
101	Improper level of packaging and packing (see procurement documents).	1.0
102	Incorrect or illegible marking.	1.0
103	Inadequate cleaning and drying.	1.5
104	Improper preservation, oil application and draining.	1.5
105	Improper containers.	1.5
106	Improperly oriented links in interior trays.	1.5
107	Improperly nested interior trays in shipping container.	1.5
108	Missing volatile corrosion inhibitor (when applicable).	1.5
109	Incorrect quantity of links per shipping container.	1.5
110	Improper closure of shipping container.	1.5
111	Even layers of links not nesting in lower layers.	1.5
Minor		
201	Workmanship	4.0

4.5.3 Testing

4.5.3.1 Material testing. Test required by applicable drawings and specifications shall be performed on materials used in the manufacture of links. The contractor shall furnish the Government representative certified statement of findings of chemical analyses and reports of physical test as may be required for each lot or each material used. The contractor shall also furnish the Government representative certification that the material used in packaging conforms to the applicable specifications.

4.5.3.2 Accelerated corrosion. Accelerated corrosion testing of links shall be made in accordance with applicable drawings and specifications.

4.5.3.3 Hardness testing. Sample links shall be taken from each heat treat lot (see 4.5.1.1) in accordance with MIL-STD-105, using inspection level S-4 and AQL of 0.40 percent defective. The links shall be tested for hardness requirements using ASTM method E18.

4.5.3.4 Carburization or decarburization. A sample of 5 links shall be selected at random from each heat treat lot (see 4.5.1.1) and tested for carburization or decarburization using test method specified in 4.6.1. Failure of any loop to meet the carburization or decarburization requirement (see 3.2.1.1) shall be cause for rejection of the heat treat lot.

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4.5.3.5 Free hinging testing. A sample of 800 links from each inspection lot shall be tested for free hinging requirements using test method specified in 4.6.2. If one or more links fail to meet the requirements it shall be cause for rejection of the lot.

4.5.3.6 Cartridge linking testing. Sample links shall be taken from each inspection lot in accordance with MIL-STD-105 and size code letter N, regardless of the quantity in the inspection lot and tested for cartridge linking using the test method specified in 4.6.3. Acceptance shall be based on an AQL 0.40 percent defective.

4.5.3.7 Cartridge stripping testing. Sample links shall be taken from each inspection lot in accordance with MIL-STD-105 and size code letter N, regardless of the quantity in the inspection lot and tested for cartridge stripping using the test method specified in 4.6.4. Acceptance shall be based on an AQL of 0.40 percent defective.

4.5.3.8 Tab radii testing. A sample of three links taken from each lot shall be tested using test method specified in 4.6.5. Failure of any link to meet the requirements shall cause rejection of the represented lot.

4.5.4 Inspection equipment

4.5.4.1 Acquisition, maintenance and disposition. Unless otherwise specified, responsibility for acquisition, maintenance and disposition of acceptance inspection and test equipment prescribed on the List of Inspection Equipment, Drawing IEL7791440, and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.4.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it must be capable of repetitive measurements to an accuracy of 10% of the total tolerance of the characteristic being inspected.

4.6 Test method

4.6.1 Carburization and decarburization. The cross section *of* the sample links (see 4.5.3.4) shall be polished, etched with a 3-5% Nital solution and examined under a microscope of not less than 500 power to determine compliance with the carburization or decarburization requirements of 3.2.1.1.

4.6.2 Free hinging test. The sample links shall be tested for free hinging requirement (see 3.2.2) by assembling the links into 20-round belts using Government standard caliber .50 dummy cartridges. Each belt shall be laid out horizontally to its full length and one end shall be drawn back upon the remainder of the belt until the belt is completely reversed. One end of the reversed belt shall then be drawn up on the remainder of the belt until the belt is returned to its original position.

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4.6.3 Cartridge linking test. The sample links shall be tested for cartridge linking requirement (see 3.2.3) using test fixture F7797437. Government standard caliber .50 dummy cartridges shall be partially inserted by hand and loaded as an assembly into test fixture. The load shall be applied as specified in 3.2.3 until the link tang overlaps the cartridge case rim. The maximum force required to insert the cartridge in each link tested shall be recorded as the cartridge linking force (see 6.1.2).

4.6.4 Cartridge stripping test. The sample links shall be tested for cartridge stripping requirement (see 3.2.4) using test fixture F7797437. Government standard caliber .50 dummy cartridges shall be inserted in the test fixture with the head of the cartridge toward the force indicator. The load shall be applied as specified in 3.2.4 until the cartridge is completely stripped from the rear loop of the link. The maximum load required to strip the cartridge from each link shall be recorded as the cartridge stripping force (see 6.1.2).

4.6.5 Tab radii test. Impression of tab radii area of each sample link shall be taken using casting material "Cerrolow 117" or equal and "Sta-warm" Electric Company's melting pot, type 6, serial no. 11134 or equal. The impression shall be measured to determine compliance with applicable requirements (see 3.2.5). Equipment used shall be subject to Government approval.

4.6.6 Functioning test. A sample of 250 links taken from the first article shall be tested by the testing agency for functioning requirement (see 3.2.6) by assembling the links into belts using Government standard caliber .50 ammunition and fired in a M85 machine gun using the following test methods.

4.6.6.1 Low rate of fire. A sample of 200 links shall be assembled into 20-round belts. Each belt shall be test fired at low rate of fire in one continuous burst.

4.6.6.2 High rate of fire. A sample of 50 links shall be assembled into 25-round belts. Each belt shall be test fired at high rate of fire in one continuous burst.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. A pilot pack shall be forwarded in accordance with the contract. Pilot pack shall be packaged to the level of packaging specified in the contract and packed level C in accordance with the requirements of Packaging Data Sheet P7791440.

5.2 Preparation for delivery shall be in accordance with Packaging Data Sheet P7791440 using the levels of preservation, packaging and packing specified in the contract (see 6.1).

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6. NOTES

6.1 Ordering data. Procurement documents should specify:

6.1.1 Procurement requirements.

a. Title, number and date of this specification.
b. List of drawings and specifications pertinent to the link, showing applicable revision dates.
c. Requirements for submission of first article by the contractor (see 3.1).
d. Index of Inspection Equipment Lists pertinent to the link shall show applicable revision. dates.
e. Responsibilities for acquisition, maintenance and disposition of acceptance inspection equipment if other than specified (see 4.5.4.1).
f. Place of final inspection and acceptance.
g. Requirements for submission of pilot pack (see 3.1 and 5.1).
h. Selection of applicable levels of preservation, packaging and packing (see 5.2).
i. Responsibility for furnishing dummy cartridges.
j. Disposition instructions for link test samples.

6.1.2 Contract data requirements. Data specified in paragraphs 4.6.3 and 4.6.4 will be listed directly on a DD Form 1423 incorporated into the contract.

6.1.3 Management control requirements. When warranted, the contract should specify the application of MIL-I-45208 on the Management Control Summary List, DD Form 1660.

6.2 When action by a testing agency is required, work programing will be effected with the testing agency at the earliest practicable date.

Custodian:
Army - WC

Preparing activity:
Army - WC

Project number:
1305-A632

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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